

DYNAMIC PORTFOLIO ALLOCATION USING DEEP REINFORCEMENT LEARNING

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MENG ELECTRICAL & COMPUTER ENGINEERING





WEALTH IS THE ABILITY TO FULLY EXPERIENCE LIFE.

-HENRY DAVID THOREAU

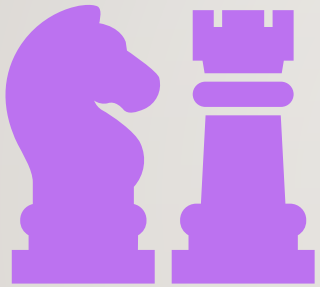




WEALTH ~~IS~~ GIVES YOU THE ABILITY TO FULLY EXPERIENCE LIFE.

-MODIFIED BY TAIMOOR SHAHZAD

BACKGROUND



Traditional approaches rely on fixed strategies, ignoring short-term patterns.



Financial portfolio management requires dynamically adjusting asset allocations.



GOAL

USE DRL TO LEARN AN ADAPTIVE, RISK-AWARE ASSET ALLOCATION POLICY.

3 PROBLEM FORMULATION

01

Agent:

DRL-based dynamic portfolio allocator

02

Environment:

Simulated portfolio return environment based on real financial market dynamics

03

State Space:

Flattened vector of past "window_size" returns for each asset (continuous)

4 PROBLEM FORMULATION – CONT'D

04

Action Space:

Continuous allocation vector (normalized to sum to 1)

05

Reward:

Weighted blend of Sharpe ratio and daily return, penalized for turnover

06

Transitions:

Daily steps based on actions taken and real return outcomes

5 PROBLEM CLASSIFICATION



Episodic (Fixed i.e. 180-day trading episodes)



Stochastic (Returns drawn from real market data)

Episode start points randomly sampled



State Space (Continuous & Finite But Large)

(window_size x num_assets)

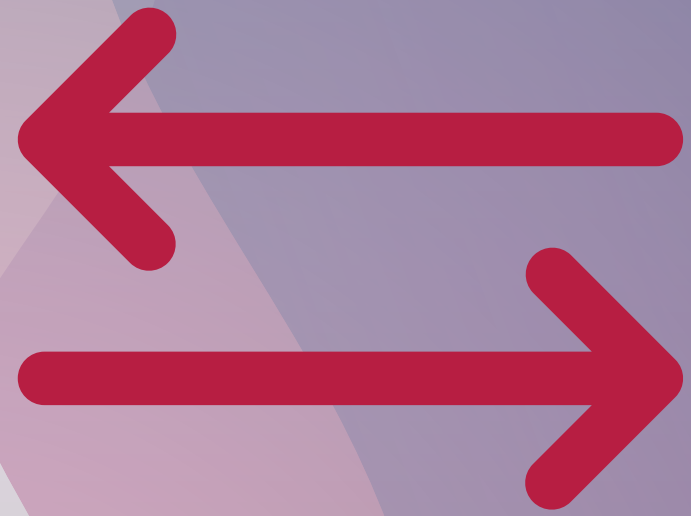


Action Space (Continuous & Infinite)

(Allocation vector sums to 1 across assets)

6 TRANSITION DYNAMICS

- Next state is a time-shifted rolling window of normalized log returns.
- Environment advances by one time step (one trading day), and the state reflects the most recent "window_size" days of returns.
- Next state depends solely on the current state and the action taken
- Satisfies Markovian Property !



7 REWARD FUNCTION FORMULATION

- **Full Reward Calculation :**

$$\text{Reward}_t = \gamma_t \cdot SR_t + (1 - \gamma_t) \cdot R_t - \lambda_{\text{tc}} \cdot \text{Turnover}_t - \lambda_{\text{pen}} \cdot \text{AllocationPenalty}_t$$

γ_t is dynamic blend between Sharpe and Return

SR_t: rolling Sharpe Ratio over recent returns

R_t: immediate portfolio return

$\lambda_{\text{tc}} = 0.001, \lambda_{\text{pen}} = 0.1$

REWARD FUNCTION FORMULATION – CONT'D

- **Sharpe Ratio:** Measure of risk-adjusted return

$$\text{Sharpe Ratio} = \frac{\mathbb{E}[R] - R_f}{\sigma[R]}$$

$\mathbb{E}[R]$ = expected portfolio return

R_f = risk-free rate (often assumed to be 0 in short-term applications)

$\sigma[R]$ = standard deviation of returns (risk)

- **Sample Turnover:** Change in portfolio allocation from the previous time step

$$\text{Turnover}_t = \sum_{i=1}^{n} \left| a_t^{(i)} - w_{t-1}^{(i)} \right|$$

$a_t^{(i)}$ = action weight for asset i

$w_{t-1}^{(i)}$ = previous weight for asset i

- **Daily Return:**

$$R_t = \mathbf{a}_t^\top \cdot \mathbf{r}_{t+1}$$

9 REWARD FUNCTION FORMULATION – CONT'D

- **Allocation Change Penalty :** Using a quadratic penalty

$$\text{Penalty}_t = \lambda_{\text{pen}} \cdot \sum_{i=1}^{n} (a_t^{(i)} - w_{t-1}^{(i)})^2$$

- **Full Reward Calculation :**

$$\text{Reward}_t = \gamma_t \cdot SR_t + (1 - \gamma_t) \cdot R_t - \lambda_{\text{tc}} \cdot \text{Turnover}_t - \lambda_{\text{pen}} \cdot \text{AllocationPenalty}_t$$

γ_t is dynamic blend between Sharpe and Return

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TRANSITION DYNAMICS SETUP

- **Time step:** 1 day
- **Window size:** 3 Trading Days (for simplicity)
- **Number of assets:** 2 (AST1, AST2)

TRANSITION DYNAMICS EXAMPLE

State

	AST 1	AST 2
D1	0.01	-0.02
D2	-0.03	0.01
D3	0.02	0.00

St = [0.01 -0.02 -0.03 0.01 0.02 0.00]

Action

Allocate weights to each asset (must sum to 1)

At = [0.6 0.4]

Environment Step

Return at t+1 (Day 4)
R_{t+1} = [0.015, -0.005]

Portfolio return:
p_ret = 0.6 × 0.015 + 0.4 × (-0.005) = 0.009

Reward (R_t): A blend of Sharpe ratio (from past returns) and this "p_ret", minus penalties.

Next State

	AST 1	AST 2
D2	-0.03	0.01
D3	0.02	0.00
D4	0.015	-0.005

St+1 = [-0.03 0.01 0.02 0.00 0.015 -0.005]

IS OUR REWARD CONSISTENT WITH THE GOAL ?

- **Adaptive Allocation:** Agent chooses a new portfolio allocation at every step
- **Risk-Aware Behavior:** Inclusion of Sharpe Ratio in reward function balances mean return and volatility (risk)
- **Asset Allocation Policy:** Learned policy is a mapping from states (recent return patterns) to allocations



APPLICABLE DRL ALGORITHMS

- Tabular methods (SARSA, Q-learning) can't be used because of large/continuous state/action space
- Starting Point: Recurrent PPO (Policy Proximal Optimization)
- Alternatives: Soft Actor Critic (SAC), TD3

WHY R-PPO ?

HANDLES LARGE CONTINUOUS STATE/ACTION SPACE

SUPPORTS PARTIAL OBSERVABILITY VIA LSTM

STABLE AND SAMPLE EFFICIENT

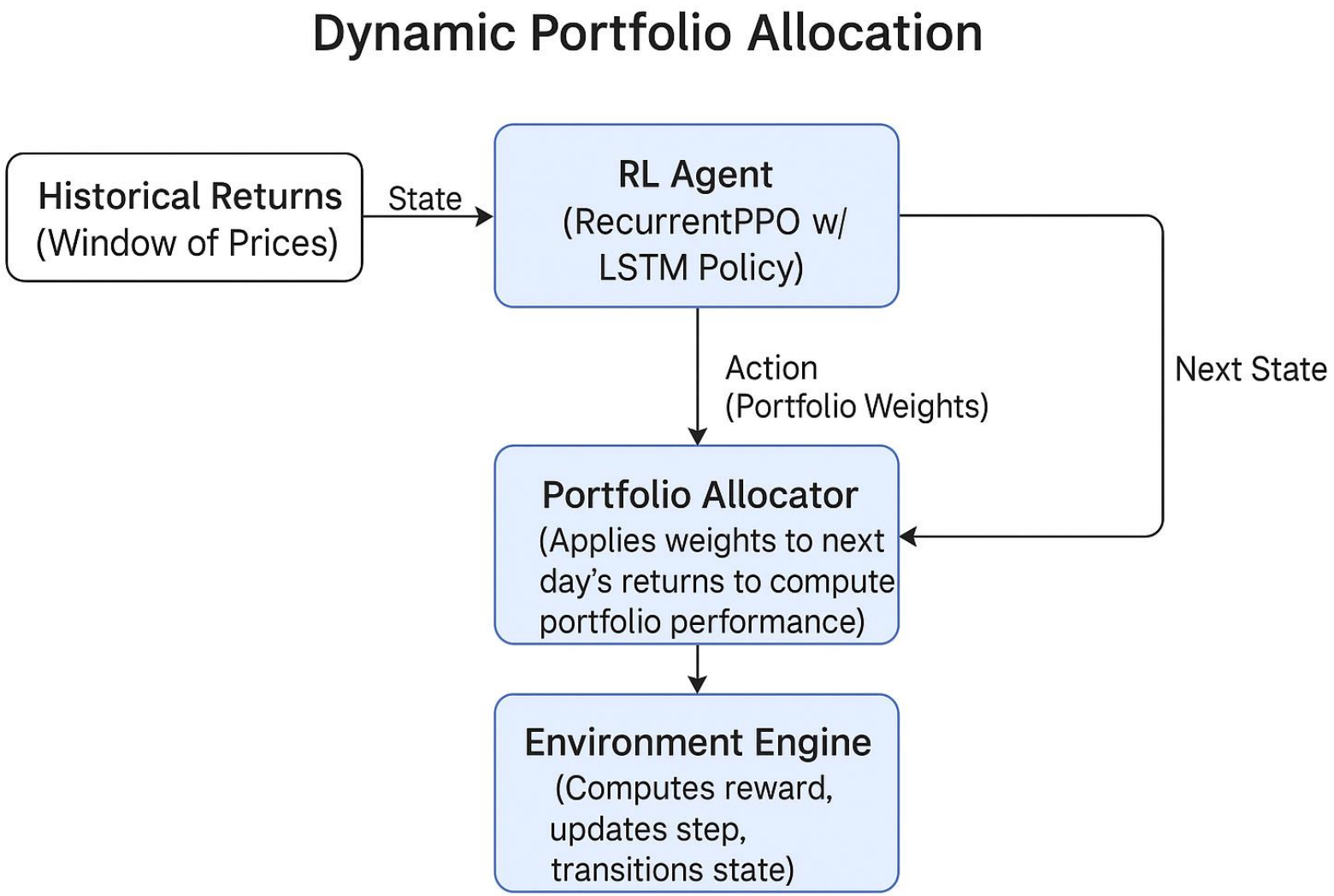
RECURRENT PPO VS TRADITIONAL METHODS

PROS	CONS
• Learns adaptive policies • Captures temporal patterns using LSTM • Direct optimization of risk-adjusted returns	• Requires extensive hyperparameter tuning • Hard to interpret learned policies • Computationally expensive

PROS	CONS
• Simple and interpretable models (MVO, Equal Weights) • Less data-hungry, easy to implement • Explainable to stakeholders	• Static or periodic allocation • Assumes stationary distributions (e.g., fixed covariance) • May perform poorly in volatile or regime-shifting markets

THANK YOU

ARCHITECTURE DIAGRAM



8 TRAINING SETUP - PPO

- Dataset: Daily Closing of 5 major assets (2015 – 2024) - Yfinance
- Splits: Train (2015 – 2020), Validate (2021 – 2023), Test (2024)
- Algorithm Used: Recurrent PPO (Policy Proximal Optimization)
- Alternatives: Soft Actor Critic (SAC), TD3

KEY HYPERPARAMETERS

Episode Length	Window Size	Reward Blend	LSTM PPO	Learning Rate	Training Rounds
180	25	Sharpe: 90%, Return: 10%	N_steps = 128, Batch Size = 64	0.0001	10

FUTURE WORK

- Include macroeconomic indicators as inputs

- Experiment with ensemble policies or SAC

- Online learning for streaming market data

- Optimize reward design and transaction penalties

OBSERVATIONS & INSIGHTS

- Learned allocation adapts well to unseen data
- Recurrent PPO captures temporal dynamics
- Dynamic reward blending helped reduce variance early in training